

Stochastic integration with respect to Lévy processes

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Setup

Definition. For a Lévy process $L = (L_t)_{t \geq 0}$, *exponential functional* of L is defined as

$$A_\infty(L) := \lim_{q \rightarrow +\infty} \int_0^q e^{-L_t} dt.$$

Integro - differential equation (Carmona, Petite, Yor, 2003). Let

$$L_t = \gamma t + \sigma W_t + (\tau_{t+} + \tau_{t-}), \quad t \geq 0,$$

where $\gamma \in \mathbb{R}$, $\sigma \in \mathbb{R}_+$, and τ_+ and $-\tau_-$ are subordinators with Lévy measures $\nu_\pm$. Then under assumptions $E[|L_1|] < \infty$, $E[L_1] < 0$, $\int_{\mathbb{R}} \bar{\nu}_\pm(x) dx < \infty$, where $\bar{\nu}_+(x) := \nu_+(x, \infty)$, $x > 0$ and $\bar{\nu}_-(x) := \nu_-(-\infty, x)$, $x < 0$, are tail integrals, r.v. $A_\infty(L)$ is infinitely divisible and absolutely continuous, and its density is a solution of the equation

$$\begin{aligned} -\frac{\sigma^2}{2} \frac{d}{dx} (x^2 p(x)) + \left(\left(\frac{\sigma^2}{2} - \gamma \right) x + 1 \right) p(x) \\ = \int_x^\infty \bar{\nu}_+(\log(u/x)) p(u) du - \int_0^x \bar{\nu}_-(\log(u/x)) p(u) du. \end{aligned}$$

Finiteness condition (Bertoin & Yor, 2005)

$$A_\infty(L) < +\infty \quad \text{a.s.} \quad \Longleftrightarrow \quad \lim_{t \rightarrow +\infty} L_t = +\infty \quad \text{a.s.}$$

Motivation 1: Carousel systems

$U_1, \dots, U_n$ - i.i.d. $\sim$ Uniform $[0, 1]$; $U_0 = 0$ - starting point; $U_{n+1} = 1$.

$D_i = U_{(i)} - U_{(i-1)}$, $1 \leq i \leq n+1$ - uniform spacings,

$$\left(D_1, D_2, \dots, D_n \right) \stackrel{d}{=} \left(\frac{S_1}{S_{n+1}}, \frac{S_2}{S_{n+1}}, \dots, \frac{S_n}{S_{n+1}} \right),$$

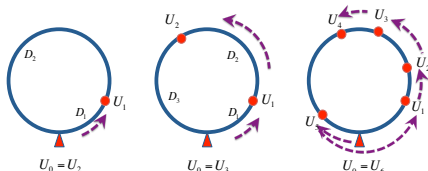
where $S_k = e_1 + \dots + e_k$, and e_i - i.i.d. $\sim \text{Exp}(1)$.

T_n - travel time under the nearest item strategy. *Litvak, Adan, 2001*:

$$T_n \stackrel{d}{=} \sum_{i=1}^n \min(D_1 + \dots + D_i, D_{i+1}) \stackrel{d}{=} \sum_{i=1}^n \left(1 - \frac{1}{2^i}\right) D_i,$$

In particular, $T_1 \stackrel{d}{=} \frac{1}{2} D_1$, $T_2 \stackrel{d}{=} \frac{1}{2} D_1 + \frac{3}{4} D_2$.

$$(n+1)(1 - T_n) \rightarrow \sum_{j=1}^{\infty} \frac{e_j}{2^{j-1}} \stackrel{d}{=} \int_0^{\infty} e^{-\log(2) * N_t} dt, \quad n \rightarrow \infty.$$



Motivation 2: Transmission control protocol (TCP)

Additive-increase multiplicative-decrease (AIMD) algorithm

$$P_{n+1} \rightarrow \begin{cases} P_n + 1, & \text{no loss among } P_n \text{ packets;} \\ \max\{\lfloor \delta P_n \rfloor, 1\}, & \text{otherwise.} \end{cases}$$

Uncorrelated loss process: each packet has a probability $1 - e^{-\alpha}$ of being lost; let the losses occur at times t_n , then $(t_{n+1} - t_n) - \text{i.i.d} \sim \text{Geom}(e^{-\alpha})$. Therefore, as $\alpha \rightarrow 0$, the loss process (number of losses to time t) can be described by a Poisson process with intensity α .

Correlated loss process: losses occur at $t_n + \tau_n^j$, where $\tau_n^1, \dots, \tau_n^J - \text{i.i.d.}$. $V_n = P_{t_n + \tau_n^{(J)}}$ - congestion window size at the end of group of losses - is the Markov chain

$$\text{Inv. d. of } \{\sqrt{\alpha} V_n\} \xrightarrow{\alpha \rightarrow 0} \text{Inv. d. of } \{\bar{V}_n\} =: \bar{V}_\infty.$$

Theorem (Guillemin, Robert, Zwart, 2004): $I \stackrel{d}{=} \bar{V}_\infty^2/2 + e_1$ satisfies

$$I \stackrel{d}{=} \delta^{2\tau_1} I + e_1, \quad \tau_1 \stackrel{d}{=} \tau_1^1, \quad e_1 \sim \text{Exp}(1).$$

$$I = \sum_{n=0}^{\infty} \delta^{2(\tau_1 + \dots + \tau_n)} e_n = \int_0^{\infty} e^{-CPP(t)} dt, \quad CPP(t) := (-2 \ln \delta) \sum_{k=1}^{N_t} \tau_k.$$

Motivation 3: Self - similar fragmentations

$s_i > 0$, $s_1 + s_2 + s_3 + \dots = 1$ and $(s_1, s_2, s_3, \dots) \sim \mathcal{L}$ - dislocation law.

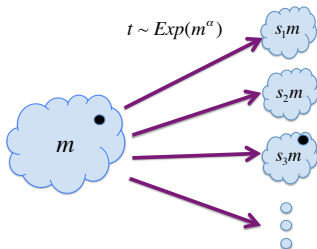
χ_t - mass of the fragment obtaining the tagged point.

Theorem (Bertoin & Yor, 2005): $X_t = 1/\chi_t$ - self-similar Markov process with index α and $t^{-1/\alpha} X_t \rightarrow \mu_\alpha$ as $t \rightarrow \infty$, where with some Lévy process L ,

$$\int_{\mathbb{R}_+} f(y) \mu_\alpha(dy) = \frac{1}{m \mathbb{E}[L_1]} \mathbb{E} \left[\frac{1}{A_\infty(\alpha L)} f \left(\frac{1}{A_\infty(\alpha L)} \right) \right].$$

The Lévy measure of L is given by $\nu(dx) = e^{-x} \sum_{j=1}^{\infty} \mathbb{P}\{-\log(s_j) \in dx\}$.

Applications: astrophysics (stellar fragments), crystallography (breaking of crystals), computer science (fragmentation of a hard drive).



Motivation 4: Econometrics

ARCH(1) model (AutoRegressive Conditionally Heteroscedastic) [*Engle, 1982*]

$$Y_n = \sigma_n \varepsilon_n, \quad \sigma_n^2 = a + b Y_{n-1}^2,$$

where ε_n - white noise process (i.i.d., $E\varepsilon_n = 0$, $\text{Var } \varepsilon_n = 1$).

GARCH(1,1) model (Generalized ARCH) [*Bollerslev, 1986*]

$$Y_n = \sigma_n \varepsilon_n, \quad \sigma_n^2 = a + b Y_{n-1}^2 + c \sigma_{n-1}^2.$$

COGARCH (COntinuous-time GARCH) [*Klüppelberg, Lindner, Maller, 2004*]

Idea: replace ε_n by the increments of a Lévy process X_t .

$$Y_t = \int \sigma_t dX_t, \quad \sigma_t^2 = e^{-L_t} \left(\sigma_0^2 + a \int_{(0,t]} e^{L_s} ds \right),$$

$$L_t = t \log c + \sum_{0 < s \leq t} \log \left(1 + \frac{c}{b} (\Delta L_s) \right).$$

$A_\infty(L)$ - invariant measure of the Generalized Ornstein-Uhlenbeck process σ_t :

$$X_t = e^{-L_t} \left(X_0 + \int_0^t e^{L_u} du \right) \rightarrow \int_0^\infty e^{-L_u} du, \quad t \rightarrow \infty.$$

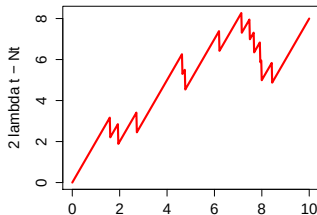
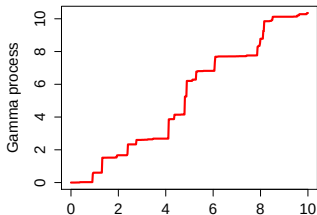
Relation to subordinators

Finiteness condition:

$$A_\infty = \int_0^{+\infty} e^{-L_t} dt < +\infty \quad \text{a.s.} \quad \Longleftrightarrow \quad \lim_{t \rightarrow +\infty} L_t = +\infty, \quad \text{a.s.}$$

Examples:

1. $L_t = \tau_t$ - a subordinator (non-decreasing Lévy process).
2. $L_t = 2\lambda t - N_t$, where N_t is a Poisson process with intensity λ .



Setup

Model:

$$A_\infty = \int_0^{+\infty} e^{-L_t} dt, \quad \text{where } L_t = \gamma t + \mathcal{T}_t,$$

where $\gamma > 0$ and $\mathcal{T}_t$ is a pure subordinator with finite Lévy measure, i.e.,

$$\nu(\mathbb{R}_-) = 0, \quad \lambda := \nu(\mathbb{R}_+) < \infty.$$

Data: Suppose that observations of the generalized OU process

$$X_t = e^{-L_t} \left(X_0 + \int_0^t e^{L_u} du \right)$$

at time points $t_1, \dots, t_n$ are given.

The aims:

- (1) to estimate γ, λ ; (2) to estimate the Lévy measure ν .

Main idea

Recursive formula (*Kuznetsov, A., Pardo, J.C. and Savov, V., 2011*):

$$\mathbb{E}[A_\infty^{s-1}] = \frac{\varphi(s)}{s} \mathbb{E}[A_\infty^s],$$

where $\varphi(s)$ is a Laplace exponent of the process L , i.e.,

$$\varphi(s) := -\log \mathbb{E}[e^{-sL_1}],$$

and s is any number from half-plane $\Upsilon := \{s \in \mathbb{C} : \operatorname{Re}(s) > 0\}$.

Laplace exponent:

$$\varphi(u + iv) = \lambda + \gamma(u + iv) - \int_0^{+\infty} e^{-ivx} \nu^*(dx),$$

where $\nu^*(dx) = e^{-ux} \nu(dx)$, and $u + iv \in \Upsilon$.

Algorithm

Consider the set $\{u + iv \in \mathbb{C} : u > 1 - \text{fixed}, v \in [\varepsilon V_n, V_n] - \text{varies}\} \subset \Upsilon$

Take the values $v_1, \dots, v_M$ on the equidistant grid on $[\varepsilon V_n, V_n]$ with a step Δ .

$$1 \quad \hat{E}[X^s] := \text{mean}_k(X_{t_n}^s) \quad \text{for } s = u + iv_m \text{ and } s = (u-1) + iv_m.$$

$$2 \quad \hat{\varphi}(s) := s \hat{E}[X^{s-1}] / \hat{E}[X^s], \quad \text{for } s = u + iv_m, \quad m = 1..M.$$

$$3 \quad \hat{\gamma}_n := \arg \min_{\gamma} \sum_{m=1}^M w_n(v_m) \left(\text{Im } \hat{\varphi}(u + iv_m) - \gamma v_m \right)^2$$

Motivation: $\text{Im } \varphi(u + iv) = \gamma v - \text{Im } \mathcal{F}_{\nu^*}(-v).$

$$4 \quad \hat{\lambda}_n := \text{mean}_m(\text{Re } \hat{\varphi}(u + iv_m)) - \hat{\gamma}_n u.$$

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$$5 \quad \hat{\mathcal{F}}_{\nu^*}(-v_m) = \hat{\lambda}_n + \hat{\gamma}_n(u + iv_m) - \hat{\varphi}(u + iv_m), \quad m = 1..M.$$

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$$6 \quad \hat{\nu}(x) = e^{ux} \frac{\Delta}{2\pi} \sum_{m=1}^M e^{iv_m x} \hat{\mathcal{F}}_{\nu^*}(-v_m) \mathcal{K}(v_m h).$$

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Simulation study

$$A_{\infty}(L) = \int_0^{\infty} e^{-L_t} dt, \quad L_t = \gamma t + \mathcal{T}_t,$$

1. $\mathcal{T}_t$ has a Lévy density

$$\nu(x) = ab \exp\{-bx\} I\{x > 0\}, \quad a, b > 0.$$

In this case, $A_{\infty}(L) \stackrel{d}{=} B(b+1, a/\gamma)/\gamma$.

2. $\mathcal{T}_t$ is a compound Poisson process,

$$\mathcal{T}_t = -(\log q) \sum_{k=1}^{N_t} \eta_k,$$

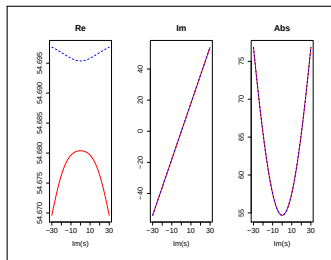
where $q \in (0, 1)$, η_k - i.i.d. $\sim \mathcal{N}_{\alpha}(0, 1)$ truncated on (α, ∞) , $\alpha \geq 0$.

$$A_{\infty}(\mathcal{T}) = \int_0^{\infty} e^{-\mathcal{T}_s} ds = \sum_{n=0}^{\infty} q^n (T_{n+1} - T_n).$$

Steps 1-2: estimation of the Laplace exponent

Estimate: $\hat{\varphi}(s) = s \frac{\hat{E}[A_{\infty}^{s-1}]}{\hat{E}[A_{\infty}^s]}, \quad s = u + iv, \quad u = 30, \quad v = 0..15.$

True value: $\varphi(s) = s \left(c + \frac{a}{b+s} \right).$

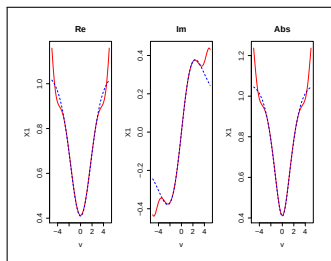


Steps 1-2: estimation of the Laplace exponent

Estimate: $\hat{\varphi}(s) = s \frac{\hat{E}[A_\infty^{s-1}]}{\hat{E}[A_\infty^s]}, \quad s = u + iv, \quad u = 30, v = 0..15.$

True value: $\varphi(s) = \lambda \left[1 - \frac{1 - F(\alpha + (\log q)s)}{1 - F(\alpha)} \exp \left\{ -\frac{(\log q)^2 s^2}{2} \right\} \right],$

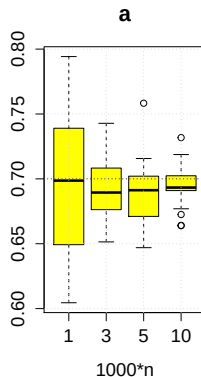
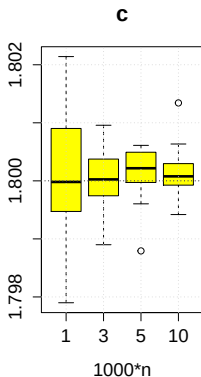
where the function $F(\cdot)$ in the complex point z can be calculated from the error function $F(z) := \frac{1}{2} \left(\operatorname{erf} \left(\frac{z}{\sqrt{2}} \right) + 1 \right)$, where $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds$.



Steps 3-4: estimation of the parameters c and a

$$\hat{\gamma}_n = \text{regr}(\text{Im } \hat{\varphi}(u + iv_m) \sim v_m),$$

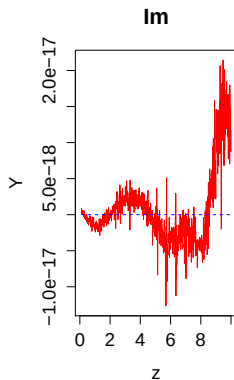
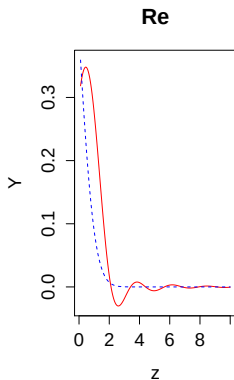
$$\hat{\lambda}_n = \text{mean}(\text{Re } \hat{\varphi}(u + iv_m)) - \hat{\gamma}_n u.$$



Steps 5-6: estimation of the Fourier transf. and the measure

$$\hat{\mathcal{F}}_{\nu^*}(-v_m) = \hat{\lambda}_n + \hat{\gamma}_n(u + iv_m) - \hat{\varphi}(u + iv_m), \quad m = 1..M.$$

$$\hat{\nu}(x) = e^{ux} \frac{\Delta}{2\pi} \sum_{m=1}^M e^{iv_m x} \hat{\mathcal{F}}_{\nu^*}(-v_m) \mathcal{K}(v_m h).$$



Theoretical study: main difficulty

Exponential inequality for i.i.d. (*Hoeffding's inequality, 1963*)

Let $X_1, X_2, \dots$ are centered, i.i.d, and $\exists M : |X_1| \leq M$ a.s. Then for any $\zeta > 0$,

$$\mathbb{P}\left\{\sum_{i=1}^n X_i \leq \zeta\right\} \geq 1 - 2 \exp\left\{-\frac{\zeta^2}{2nM^2}\right\}.$$

Dependence concept: α -mixing. For sigma-algebras $\mathcal{B}$ and $\mathcal{C}$,

$$\alpha(\mathcal{B}, \mathcal{C}) := \sup_{B \in \mathcal{B}, C \in \mathcal{C}} |\mathbb{P}\{B \cap C\} - \mathbb{P}\{B\} \mathbb{P}\{C\}|$$

A sequence $X_1, X_2, \dots$ is said to be exponentially α -mixing, if

$$\alpha_k := \sup_{t \in \mathbb{Z}} \alpha(\sigma(X_s, s \leq t), \sigma(X_s, s \geq t+k)) \leq e^{-k\alpha^*} \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Exponential inequalities (*Merlevède, Peligrad, Rio, 2009*)

Let $X_1, X_2, \dots$ are centered, exponentially α -mixing, and $|X_k| \leq M, \forall k$ a.s. Then

$$\mathbb{P}\left\{\sum_{i=1}^n X_i \leq \zeta\right\} \geq 1 - \exp\left\{-\frac{C\zeta^2}{nv^2 + M^2 + M\zeta \log^2(n)}\right\}$$

for all $\zeta > 0$ and $n \geq 4$, where $v^2 = \sup_i \left(\mathbb{E}[X_i]^2 + 2 \sum_{j \geq i} \text{Cov}(X_i, X_j) \right)$.

Mixing for multidimensional diffusions with jumps: *Masuda (2007)*

Quality of $\hat{\gamma}_n$ and $\hat{\lambda}_n$

Rates of convergence

For $s \in \mathbb{N} \cup \{0\}$ and $R > 0$, let $\mathcal{G}(s, R)$ denote the set of all Lévy triplets $(\gamma, 0, \nu)$, such that ν is supported on $\mathbb{R}_+$ and

$$\max \left\{ \nu(\mathbb{R}_+), \int_{\mathbb{R}} |v|^{2s} |\mathcal{F}[\nu^*](v)|^2 dv \right\} \leq R.$$

where $\nu^*(dx) = e^{-u^\circ x} \nu(dx)$. Let A_∞ belong to the class

$$\mathcal{P}(\beta) := \left\{ p : \exists L > 0 \text{ s.t. } \liminf_{|v| \rightarrow \infty} \left[|v|^\beta |\mathcal{M}[p](u^\circ + iv)| \right] \geq L \right\}.$$

Suppose that the sequence $X_0, X_1, \dots, X_n$ is exponentially α -mixing and strictly stationary. Under some choice of V_n ,

$$\mathbb{E} \left[|\hat{\gamma}_n - \gamma|^2 \right] \lesssim n^{-2(s+2)/(2\beta+2s+3)} \log(n),$$

$$\mathbb{E} \left[|\hat{\lambda}_n - \lambda|^2 \right] \lesssim n^{-2(s+1)/(2\beta+2s+3)} \log(n),$$

$$\mathbb{E} \left[\int_{\mathbb{R}} |\bar{\nu}_n(x) - \bar{\nu}(x)|^2 dx \right] \lesssim n^{-2s/(2\beta+2s+3)}.$$

Why do we need complex numbers in this task at all?

Inverse Laplace transform (Bromwich integral):

$$\nu(x) = \frac{1}{2\pi i} \int_{\Omega} e^{sx} \mathcal{L}(s) ds, \quad \Omega := \{\operatorname{Re}(s) = \gamma\},$$

where γ is greater than the real part of all singularities of $\mathcal{L}(s)$.

The shortest path between two truths in the real domain passes through the complex domain (Jacques Hadamard).

Setup

Moving-average Lévy process is a stochastic integral

$$Z_t = \int_{\mathbb{R}} \mathcal{K}(t-s) dL_s,$$

- L_t is a Lévy process on $\mathbb{R}$: $L_t = L_t^1 \mathbb{I}\{t \geq 0\} - L_{(-t)}^2 \mathbb{I}\{t < 0\}$, where L_t^1 and L_t^2 are two independent copies of a Lévy pr. with triplet (γ, σ^2, ν) ,
- $\mathcal{K}(\cdot) : \mathbb{R} \rightarrow \mathbb{R}_+$ is a deterministic function.

$\mathcal{K} \in \mathcal{L}^1(\mathbb{R}) \cap \mathcal{L}^2(\mathbb{R})$, $\int_{\mathbb{R}^2} x^2 \nu(dx) < \infty$ guarantee the stationarity of Z_t :

$$\Phi(u) := \mathbb{E}[e^{iuZ_t}] = \exp\left\{\int_{\mathbb{R}} \psi(u\mathcal{K}(x)) dx\right\},$$

where $\psi(u) = t^{-1} \log(\mathbb{E}[e^{iuL_t}])$ is the characteristic exponent of L_t

$$\psi(u) = i\gamma u - \frac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R}} \left(e^{iux} - 1 - iux\mathbb{I}\{|x| \leq 1\}\right) \nu(x) dx.$$

Important: Z_t is an integral over $\mathbb{R}$.

Typical example: $\mathcal{K}(x) = |x|^r e^{-\rho|x|}$ with $r > -1/2$ and $\rho > 0$.

Moving-average process: motivation

Lévy-driven Ornstein-Uhlenbeck process:

$$Z_t = e^{-\rho t} \left(Z_0 + \int_0^t e^{\rho s} dL_s \right) \quad \text{with } \rho > 0.$$

Well-balanced Ornstein-Uhlenbeck process (*Schnurr & Woerner, 2011*):

$$K(x) = e^{-\rho|x|} \implies Z_t = \int_{\mathbb{R}} e^{-\rho|t-s|} dL_s.$$

More general set-up: $Z_t = \int_{\mathbb{R}} K(t, s) dL_s$, where, for instance,

$$K(t, s) := \frac{1}{\Gamma(1 + \gamma)} [(t - s)_+^\gamma - (-s)_+^\gamma]$$

with $\gamma \in (0, 1/2)$ (fractional Lévy process).

Barndorff-Nielsen & Schmiegel (2009), Basse & Pedersen (2009), Podolskij (2014), Barndorff-Nielsen, Benth & Veraart (2015), Basse-O'Connor & Rosiński (2016)

Related process: fractional Lévy process (FLP)

$$F_{\gamma}(t) = \int_{\mathbb{R}} \frac{1}{\Gamma(1+\gamma)} [(t-s)_{+}^{\gamma} - (-s)_{+}^{\gamma}] dL_s \quad \text{with } 0 < \gamma < 1/2.$$

1. Infinitely divisible distribution: for any $t > 0$, $F_{\gamma}(t)$ has an inf. div. distr.
2. Covariance function:

$$\text{Cov}(F_{\gamma}(t), F_{\gamma}(s)) = C_{\gamma} \left(|t|^{2\gamma+1} - |t-s|^{2\gamma+1} + |s|^{2\gamma+1} \right)$$

3. Stationary increments: $F_{\gamma}(t) - F_{\gamma}(s) \stackrel{d}{=} F_{\gamma}(t-s)$.
4. Symmetry: $F_{\gamma}(-t) = F_{\gamma}(-t) - F_{\gamma}(0) \stackrel{d}{=} F_{\gamma}(0) - F_{\gamma}(-t) = -F_{\gamma}(-t)$.

Marquardt (Bernoulli'06)

Direct approach

Statistical problem: Let $Z_t = \int_{\mathbb{R}} \mathcal{K}(t-s) dL_s$ be stationary with c.f.

$$\Phi(u) = \mathbb{E}[e^{iuZ_t}] = \exp\left\{\int_{\mathbb{R}} \psi(u\mathcal{K}(x)) dx\right\},$$

where $\psi(u)$ is the characteristic exponent of L_t .

Aim: estimate $\nu(x)$ from $Z_0, Z_{\Delta}, \dots, Z_{n\Delta}$ with fixed $\Delta > 0$.

Special choice of $\mathcal{K}$. Let $\mathcal{K}$ be a symmetric kernel of the form:

$$\mathcal{K}_{\alpha}(x) := (1 - \alpha|x|)^{\frac{1}{\alpha}}, \quad |x| \leq \alpha^{-1},$$

for some $\alpha \in (0, 1)$. In this case, $\mathcal{K}'_{\alpha}(x) = -\mathcal{K}_{\alpha}^{1-\alpha}(x)$.

Direct relation between ψ and Φ .

$$\psi(u) = \frac{1}{2} u^{1-\alpha} (u^{\alpha} \log(\Phi(u)))' = \frac{1}{2} \left(\alpha \log(\Phi(u)) + u \frac{\Phi'(u)}{\Phi(u)} \right).$$

Direct approach: estimation procedure

1. Estimation of Φ, ψ .

$$\Phi_n(u) := \frac{1}{n} \sum_{j=1}^n e^{iuZ_j\Delta}, \quad \psi_n(u) = \frac{1}{2} \left(\alpha \log(\Phi_n(u)) + u \frac{\Phi_n'(u)}{\Phi_n(u)} \right).$$

2. Estimation of σ, λ, γ . Since

$$\psi(u) = i\gamma u - \frac{1}{2}\sigma^2 u^2 - \lambda + \mathcal{F}[\nu](u),$$

we define

$$\begin{aligned} (\sigma_n^2, \lambda_n) &:= \arg \min_{(\sigma^2, \lambda)} \int_0^\infty w^{U_n}(u) (\operatorname{Re} \psi_n(u) + \sigma^2 u^2 / 2 + \lambda)^2 du \\ \gamma_n &:= \arg \min_{\gamma} \int_0^\infty w^{U_n}(u) (\operatorname{Im} \psi_n(u) - \gamma u)^2 du. \end{aligned}$$

3. Estimation of ν :

$$\nu_n(x) := \mathcal{F}^{-1} \left[\left(\psi_n(\cdot) - i\gamma_n[\cdot] + \frac{\sigma_n^2}{2} [\cdot]^2 + \lambda_n \right) w_\nu(\cdot / U_n) \right] (x), \quad x \in \mathbb{R}.$$

Direct approach: theoretical results

Consider the class

$$\mathcal{T}_s = \mathcal{T}_s(\sigma^\circ, R) = \left\{ \sigma \in (0, \sigma^\circ), \int x^2 \nu(dx) \leq R, \left\| \nu^{(s)} \right\|_\infty \leq R \right\}$$

Optimality in the minimax sense

Upper bound. If $\|\mathcal{F}(w_\sigma^1(u)/u^s)\|_{L^1} < \infty$, then it holds

$$\lim_{A \rightarrow +\infty} \overline{\lim}_{n \rightarrow +\infty} \sup_{(\gamma, \sigma, \nu) \in \mathcal{T}_s} \mathbb{P} \left\{ \left| \sigma_n^2 - \sigma^2 \right| \geq A \cdot U_n^{-(s+3)} \right\} = 0.$$

provided $U_n = \sqrt{\kappa \log(n)}$ with some constant $\kappa > 0$ depending on σ°, R .

Lower bound. For any $\sigma^\circ, R > 0$, there exists some $A > 0$ such that

$$\lim_{n \rightarrow +\infty} \inf_{\check{\sigma}_n} \sup_{(\gamma, \sigma, \nu) \in \mathcal{T}_s} \mathbb{P} \left\{ \left| \check{\sigma}_n^2 - \sigma^2 \right| \geq A \cdot (\log(n))^{-(s+3)/2} \right\} > 0,$$

where the infimum is taken over all possible estimates $\check{\sigma}_n, \check{\gamma}_n, \check{\lambda}_n$ of the parameters σ, γ, λ , and supremum - over all triplets from the class $\mathcal{T}_s$.

Mellin transform $\mathcal{M}[f](z) := \int_{\mathbb{R}_+} x^{z-1} f(x) dx, \quad z \in \mathbb{C}.$

Example: $f(x) = e^{-x}, x > 0 \implies \mathcal{M}[f](z) = \int_{\mathbb{R}_+} x^{z-1} e^{-x} dx = \Gamma(z).$

Closely related to Fourier transform $\mathcal{F}[f](u) := \int_{\mathbb{R}} e^{iux} f(x) dx, \quad u \in \mathbb{R},$
to Laplace transform $\mathcal{L}[f](s) := \int_{\mathbb{R}_+} e^{-sx} f(x) dx, \quad s \in \mathbb{C}.$

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Properties

- 1 **Convergence** follows from the properties of the Laplace transform:
the region of convergence $\mathcal{C}_f$ is
either a strip $\{z : \operatorname{Re}(z) \in \langle a, b \rangle\}$ with $a < b$
or a line $\{z : \operatorname{Re}(z) = a\}$ with $a \in \mathbb{R}.$

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Properties

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either a strip $\{z : \operatorname{Re}(z) \in \langle a, b \rangle\}$ with $a < b$
or a line $\{z : \operatorname{Re}(z) = a\}$ with $a \in \mathbb{R}.$
- 2 **Parsenval formula** (similar to Fourier transform): for any $c \in \mathcal{C}_f \cap \mathcal{C}_g$

$$\int_{\mathbb{R}_+} f(x)g(x)dx = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \mathcal{M}[f](z) \cdot \mathcal{M}[g](1-z) dz.$$

- 3 **Inverse Mellin transform** (similar to Fourier transform): for any $c \in \mathcal{C}_f$

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} x^{-z} \cdot \mathcal{M}[f](z) dz.$$

Mellin transform $\mathcal{M}[f](z) := \int_{\mathbb{R}_+} x^{z-1} f(x) dx, \quad z \in \mathbb{C}.$

4 Multiplicative convolution

$$f \star g(u) := \int_{\mathbb{R}_+} f(ux)g(x)dx \implies \mathcal{M}[f \star g](z) = \mathcal{M}[f](z) \mathcal{M}[g](1-z),$$

$$X = Y\zeta, \quad Y \perp\!\!\!\perp \zeta \implies \mathbb{E}[X^{z-1}] = \mathbb{E}[Y^{z-1}]\mathbb{E}[\zeta^{z-1}].$$

5 Superposition of Mellin and Laplace / Fourier transforms

$$\mathcal{M}[\mathcal{L}[f]](z) = \mathcal{M}[f](1-z) \cdot \Gamma(z).$$

6 Various ideas related to complex integration:

Cauchy's integral theorem, change of variable,
integration along curves in complex plane ...

Superposition of Mellin and Fourier transforms

Denote $\Lambda(u) := \log(\Phi(u))$, and consider the Mellin transform of $\Lambda''(u)$:

$$\mathcal{M}[\Lambda''](z) = -\mathcal{M}[\mathcal{F}[\bar{\nu}]](z) \cdot \int_{\mathbb{R}} (\mathcal{K}(x))^{2-z} dx, \quad \text{where } \bar{\nu}(x) = x^2 \cdot \nu(x).$$

$$\boxed{\mathcal{M}[\bar{\nu}](1-z) = \frac{\mathcal{M}[\Lambda''](1-z)}{Q(1-z)}} \quad \text{with } Q(z) := -\Gamma(z)e^{i\pi z/2} \int_{\mathbb{R}} (\mathcal{K}(x))^{2-z} dx.$$

Estimation scheme

$$1. \quad X_0, X_{\Delta}, \dots, X_{n\Delta} \longrightarrow \Phi \longrightarrow \mathcal{M}[\Lambda'']:$$

$$\mathcal{M}_n[\Lambda''](1-z) := \int_0^{U_n} \left[\frac{\Phi_n''(u)}{\Phi_n(u)} - \left(\frac{\Phi_n'(u)}{\Phi_n(u)} \right)^2 \right] u^{-z} du,$$

where $\Phi_n(u) := n^{-1} \sum_{k=1}^n \exp\{iX_{k\Delta}u\}$ and a sequence $U_n \rightarrow \infty$.

$$2. \quad \mathcal{M}[\Lambda''] \longrightarrow \mathcal{M}[\bar{\nu}] \longrightarrow \bar{\nu}:$$

$$\bar{\nu}_n(x) := \frac{1}{2\pi i} \int_{c-iV_n}^{c+iV_n} \frac{\mathcal{M}_n[\Lambda''](1-z)}{Q(1-z)} x^{-z} dz,$$

where $V_n \rightarrow \infty$ and $c \in (0, 1)$.

Assumptions

- Typical example:

$$\nu(x) = \sum_{j=1}^J a_j x^{-\eta_j-1} e^{-\lambda_j x} \cdot \mathbb{I}\{x \geq 0\}$$

with $a_j > 0, \eta_j < 1, \lambda_j > 0, \forall j$.

$$\mathcal{K}(x) = \left(\sum_{r=0}^R \beta_r |x|^r \right) e^{-\rho|x|} \quad \text{with } \beta_r \geq 0 \text{ for all } r, \text{ and } \rho > 0.$$

- The measure ν is supported on $\mathbb{R}_+$, fulfills $\int_0^1 x \nu(x) dx < \infty$, and for some $A > 0, \alpha \in (0, 1), \gamma > 0, c \in (0, 1)$, it holds

$$\max \left\{ \int_{\mathbb{R}} (1 + |y|)^\alpha |\mathcal{F}[\bar{\nu}](y)| dy, \int_{\mathbb{R}} e^{\gamma|u|} |\mathcal{M}[\bar{\nu}](c + iu)| du \right\} \leq A,$$

Moreover, assume that for some $R > 0, \int_{x>1} e^{Rx} \nu(x) dx \leq A$.

- The kernel $\mathcal{K} \in \mathcal{L}^1(\mathbb{R}) \cap \mathcal{L}^2(\mathbb{R})$ satisfies

$$\sum_{j=-\infty}^{\infty} \left| \mathcal{F}[\mathcal{K}] \left(2\pi \frac{j}{\Delta} \right) \right| \leq K^*; \quad (\mathcal{K} \star \mathcal{K})(\Delta j) \leq \kappa_0 |j|^{\kappa_1} e^{-\kappa_2 |j|}, \quad \forall j \in \mathbb{Z}$$

for some positive constants K^*, κ_0, κ_1 and κ_2 , and moreover all eigenvalues of the matrix $\mathcal{M} = \left((\mathcal{K} \star \mathcal{K})(\Delta(j-k)) \right)_{k,j \in \mathbb{Z}}$ are bounded from below and above by two finite positive constants.

Upper bounds for $\bar{\nu}_n(x)$

Let us fix $K > 0$ and denote the set

$$\mathcal{A}_K := \left\{ \max_{j=0,1,2} \left\| \frac{\Phi_n^{(j)}(u) - \Phi^{(j)}(u)}{\Phi(u)} \right\|_{U_n} \leq K\varepsilon_n \right\},$$

where for any $f : \mathbb{R} \rightarrow \mathbb{R}$, $\|f\|_{U_n} := \sup_{u \in [-U_n, U_n]} |f(u)|$, $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$, and $K\varepsilon_n (1 + \|\Psi'_\sigma\|_{U_n}) \leq 1/2$. Then on the set $\mathcal{A}_K$,

$$\sup_{x \in \mathbb{R}} \{|x|^c |\bar{\nu}_n(x) - \bar{\nu}(x)|\} \leq \frac{1}{2\pi} \int_{\{|v| \leq V_n\}} \frac{\Omega_n}{Q(v)} dv + \frac{A}{2\pi} e^{-\beta V_n},$$

where $\Omega_n \sim U_n^{1-c}$, $Q(v) = \min(|Q_1(1-c-iv)|, |Q_2(1-c-iv)|)$, $Q_j(z) := -\frac{e^{i\pi z} - e^{-i\pi z}}{e^{(-1)^j + 1} i\pi z/2} \Gamma(z) \int_{\mathbb{R}} (\mathcal{K}(x))^{2-z} dx$. Under the choice

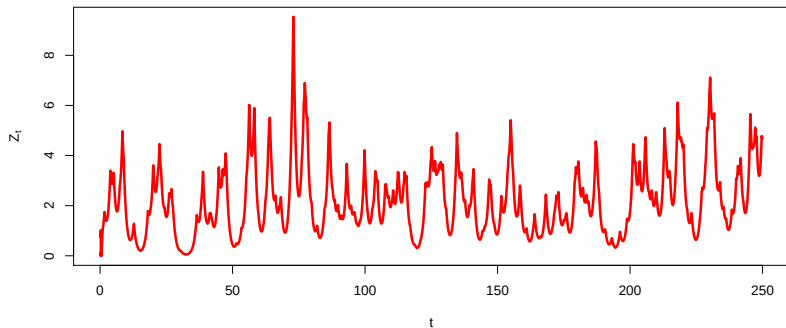
$\varepsilon_n = \sqrt{\frac{\log(n)}{n}} \cdot \exp\left(\frac{A}{2}\sigma^2 U_n^2 \int (\mathcal{K}(x))^2 dx\right)$, it holds for any $K > 0$

$$\mathbb{P}\{\mathcal{A}_K\} \geq 1 - \frac{C_1}{\sqrt{K}} \frac{\sqrt{U_n} n^{(1/4)-C_2 K^2}}{\log^{1/4}(n)}.$$

Example

Consider the integral $Z_t := \int_{\mathbb{R}} \mathcal{K}(t-s) dL_s$, where

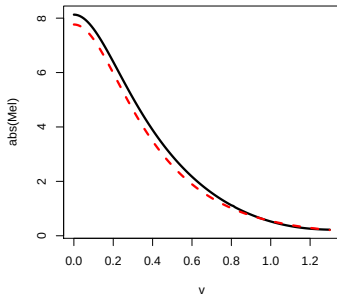
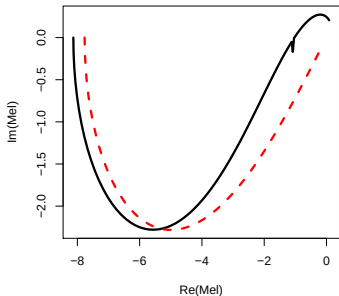
1. $\mathcal{K}(x) = e^{-|x|}, x \in \mathbb{R}$;
2. L_t is constructed from the compound Poisson process
 $L_t^{(1)} \stackrel{d}{=} L_t^{(2)} \stackrel{d}{=} \sum_{k=1}^{N_t} \xi_k$, where N_t is a Poisson process with intensity λ , and $\xi_1, \xi_2, \dots$ are independent r.v.'s with standard exponential distribution.



Estimation of $\mathcal{M}[\Lambda']$: practice

We estimate the Mellin transform $\mathcal{M}[\Lambda'](1-z) = \int_0^\infty \frac{\Phi'(u)}{\Phi(u)} u^{-z} du$ at the points $z = c + iv$, $v = -V_n..V_n$, with step $\delta = 2V_n/K$:

$$\mathcal{M}_n[\Lambda'](1-z) := i \int_0^{U_n} \left[\frac{\text{mean}(Z_{k\Delta} e^{iuZ_{k\Delta}})}{\text{mean}(e^{iuZ_{k\Delta}})} - \text{mean}(Z) e^{iu} \right] u^{-z} du + 2i\lambda\Gamma(1-z) \exp\{i\pi(1-z)/2\}.$$

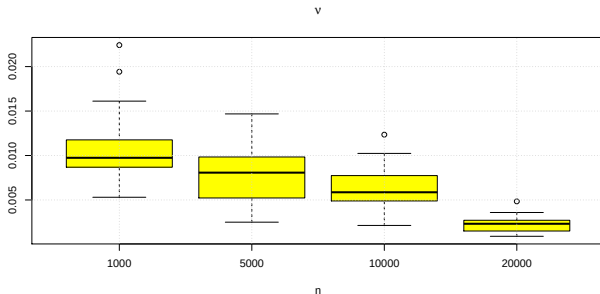


Estimation of $\nu(x)$: practice

Next, we estimate the Lévy density $\nu(x)$ by

$$\nu_n(x) := \frac{\delta}{2\pi x} \sum_{k=1}^K \operatorname{Re} \left\{ \frac{\mathcal{M}_n[\Lambda'](1 - c - iv_k)}{\tilde{Q}(1 - c - iv)} x^{-(c+iv_k)} \right\},$$

where $\tilde{Q}(z) = i\Gamma(z) \exp\{i\pi z/2\} \int_{\mathbb{R}} (K(x))^{1-z} dx$, and measure the quality in terms of $\mathcal{R}(\nu_n) = \int_1^3 (\nu_n^*(x) - \nu(x))^2 dx$.



Summary

- We consider stochastic integrals of the following two types

$$\int_{\mathbb{R}_+} e^{-L_t} dt, \quad \int_{\mathbb{R}} \mathcal{K}(t-s) dL_s.$$

- Nice math constructions with many possible applications!
- Statistical inference is possible: Fourier-based and Mellin-based methods.
- Main challenge from theoretical point of view: mixing properties.

- 1 Belomestny D., Panov V. (2015) Statistical inference for generalized Ornstein-Uhlenbeck processes. *Electronic Journal of statistics*. Vol. 9. No. 2. P. 1974–2006.
- 2 Belomestny D., Panov V., Woerner J. (2019) Low-frequency estimation of continuous-time moving average Levy processes. *Bernoulli*. Vol. 25. No. 2. P. 902–931
- 3 Belomestny D., Orlova T., Panov V. (2019) Statistical inference for moving-average Lévy-driven processes: Fourier-based approach. *Statistica Neerlandica*. Vol. 1. P. 100–117.

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Thank you for your attention.